Team 12

Limit order book system

ENTITY RELATIONSHIP DOCUMENTATION

IDENTIFICATION OF ENTITIES

1. User
2. Account
3. Session
4. VerificationToken
5. Cryptocurrency
6. Exchange
7. TradingPair
8. OrderBook
9. BidOrder
10. AskOrder
11. MarketData
12. PriceHistory
13. Trade
14. Watchlist
15. WebSocketConnection

ENTITY–ATTRIBUTE TABLE

Entity: User

Attribute	Description
id (PK)	Unique identifier for user
name	User name
email	User email (unique)
emailVerified	Email verification time
image	Profile image
createdAt	Account creation time
updatedAt	Last update time

Entity: Account

Attribute	Description
id (PK)	Account identifier
userId (FK)	References User
type	Account type
provider	OAuth provider
providerAccountId	Provider account ID
access_token	Access token
refresh_token	Refresh token
expires_at	Token expiry

Entity: Session

Attribute	Description
id (PK)	Session identifier
sessionToken	Unique session token
userId (FK)	References User
expires	Session expiry time

Entity: VerificationToken

Attribute	Description
identifier (PK)	Email or identifier
token (PK)	Verification token
expires	Expiry time

Entity: Cryptocurrency

Attribute	Description
id (PK)	Cryptocurrency ID
symbol	BTC, ETH, etc.
name	Cryptocurrency name
description	Description
logoUrl	Logo URL
rank	Market rank
createdAt	Created time
updatedAt	Updated time

Entity: Exchange

Attribute	Description
id (PK)	Exchange ID
name	Exchange name
code	Exchange code
description	Description
websocketUrl	WebSocket endpoint
apiUrl	API endpoint
isActive	Exchange status
createdAt	Created time
updatedAt	Updated time

Entity: TradingPair

Attribute	Description
id (PK)	Trading pair ID
cryptocurrencyId (FK)	References Cryptocurrency
exchangeId (FK)	References Exchange
baseCurrency	Base currency
quoteCurrency	Quote currency
pairSymbol	Pair symbol
isActive	Active status
createdAt	Created time
updatedAt	Updated time

Entity: OrderBook

Attribute	Description
id (PK)	Order book snapshot ID
tradingPairId (FK)	References TradingPair
timestamp	Snapshot time
sequenceNumber	Sequence number
checksum	Integrity checksum
createdAt	Created time

Entity: BidOrder

Attribute	Description
id (PK)	Bid order ID
orderBookId (FK)	References OrderBook
price	Bid price
size	Quantity
total	Total value
orderPosition	Position in order book
createdAt	Created time

Entity: AskOrder

Attribute	Description
id (PK)	Ask order ID
orderBookId (FK)	References OrderBook
price	Ask price
size	Quantity
total	Total value
orderPosition	Position in order book
createdAt	Created time

Entity: MarketData

Attribute	Description
id (PK)	Market data ID
tradingPairId (FK)	References TradingPair
timestamp	Time of record
currentPrice	Current price
highPrice24h	24h high
lowPrice24h	24h low
volume24h	Trading volume
spread	Bid–ask spread
createdAt	Created time

Entity: PriceHistory

Attribute	Description
id (PK)	History ID
tradingPairId (FK)	References TradingPair
timestamp	Time
interval	Time interval
openPrice	Open price
highPrice	High price
lowPrice	Low price
closePrice	Close price

Entity: Trade

Attribute	Description
id (PK)	Trade ID
tradingPairId (FK)	References TradingPair
tradeId	Exchange trade ID
timestamp	Trade time
price	Trade price
size	Trade size
side	BUY / SELL
isBuyerMaker	Maker flag
createdAt	Created time

Entity: Watchlist

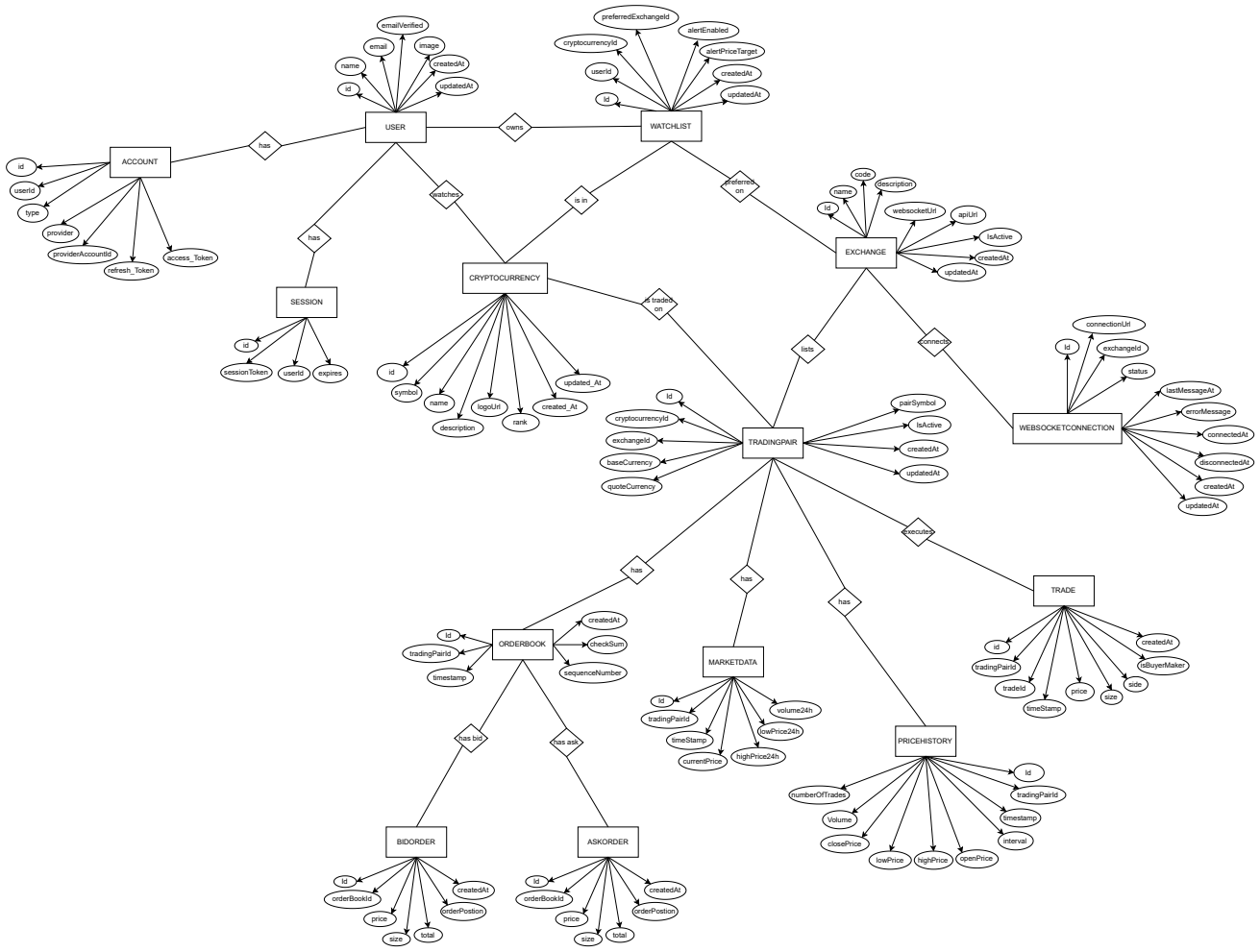
Attribute	Description
id (PK)	Watchlist ID
userId (FK)	References User
cryptocurrencyId (FK)	References Cryptocurrency
preferredExchangeId (FK)	References Exchange
notes	User notes
alertEnabled	Alert status
alertPriceTarget	Target price
createdAt	Created time
updatedAt	Updated time

Entity: WebSocketConnection

Attribute	Description
id (PK)	Connection ID
exchangeId (FK)	References Exchange
connectionUrl	WebSocket URL
status	Connection status
lastMessageAt	Last message time
errorMessage	Error details
createdAt	Created time
updatedAt	Updated time

RELATIONSHIP DESCRIPTION TABLE

From Entity	To Entity	Relationship Type	Description
User	Account	One-to-Many	User can have multiple accounts
User	Session	One-to-Many	User can have multiple sessions
User	Watchlist	One-to-Many	User tracks many cryptocurrencies
Cryptocurrency	TradingPair	One-to-Many	Crypto traded on multiple exchanges
Exchange	TradingPair	One-to-Many	Exchange lists multiple pairs
TradingPair	OrderBook	One-to-Many	Multiple snapshots over time
OrderBook	BidOrder	One-to-Many	Multiple bid orders
OrderBook	AskOrder	One-to-Many	Multiple ask orders
TradingPair	MarketData	One-to-Many	Market statistics
TradingPair	PriceHistory	One-to-Many	Historical prices
TradingPair	Trade	One-to-Many	Executed trades
User	Cryptocurrency	Many-to-Many	Implemented via Watchlist
Exchange	WebSocketConnection	One-to-Many	Connection monitoring


USER
id
name
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createdAt
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WATCHLIST
id
userId
cryptocurrencyId
preferredExchangeId
alertEnabled
alertPriceTarget
createdAt
updatedAt
has
ACCOUNT
id
userId
type
provider
providerAccountId
refresh_Token
access_Token
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SESSION
id
sessionToken
userId
expires
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CRYPTOCURRENCY
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logoUrl
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description
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is traded on
EXCHANGE
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apiUrl
isActive
createdAt
updatedAt
lists
connects
WEBSOCKETCONNECTION
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exchangeId
status
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errorMessage
connectedAt
disconnectedAt
createdAt
updatedAt
TRADINGPAIR
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cryptocurrencyId
exchangeId
baseCurrency
quoteCurrency
pairSymbol
isActive
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updatedAt
executes
TRADE
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tradeId
timeStamp
price
size
side
isBuyerMaker
createdAt
has
ORDERBOOK
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tradingPairId
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checkSum
sequenceNumber
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MARKETDATA
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tradingPairId
timeStamp
currentPrice
highPrice24h
lowPrice24h
volume24h
has
PRICEHISTORY
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interval
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lowPrice
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has ask
BIDORDER
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orderBookId
price
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createdAt
ASKORDER
id
orderBookId
price
size
total
orderPostion
createdAt